

COMPILERS QUIZ 3: EXTENDED

1 Loops

The word ``P forever'` executes the stored procedure ``P'` on top of the stack forever after popping it.

```
0 { inc dup print } forever
```

prints all integers starting from 0.

The word ``P xs foreach'` executes ``P'` with each element of ``xs'` on top of the stack one-by-one. Both, ``P'` and ``xs'` are popped in the beginning.

```
{ print } [ 2 3 5 7 ] foreach
```

prints the first four prime numbers.

2 Command-line arguments

The word ``argv'` pushes the list of command-line arguments (strings) to the stack. As usual, the first element is the name of the executable and the remaining are the arguments.

Suppose I run a script as:

```
./run.sh echo.f "Hello, World"
```

The program in the file ``echo.f'` is executed with the list ``argv'` as: the first element is the string ``echo.f'` and the second element is the string ``Hello, World'`.

The following script is an implementation of UNIX ``yes'` command.

```
{ argv len 1 > } { argv 1 nth } { "y" } if  
{ dup print } forever
```

3 Type predicates

Implement ``is-number?', `is-list?', `is-string?', `is-bool?' predicates to test whether the element on top of the stack is of corresponding type.`